KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

HP Chromebook 11 G9 EE

Solder-Free Repair Guide



Every field-replaceable repair that needs a screwdriver, not a soldering iron.
HP Chromebook 11 G9 Education Edition (Intel).

In this guide

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Built from HP's 11MK G9 EE service guide (sister model). Confirm details.
Part numbers: verify on partsurfer.hp.com before ordering.

1 Read first

This guide covers every HP Chromebook 11 G9 EE repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is in Section 6 and goes to the lead tech or the manufacturer.

How this guide was built. HP's service guide for this exact model wasn't publicly available. Procedures follow HP's guide for the 11MK G9 EE, the same G9 generation. Steps that may differ say so and have notes boxes; confirm them on your first teardown.

Built from the sister model's guide. HP's service guide for the 11 G9 EE (Intel) wasn't available. This guide uses the 11MK G9 EE guide (same G9 generation and chassis style). Part numbers differ by processor: look yours up on partsurfer.hp.com by serial number, and confirm screw counts in the notes boxes on your first teardown.

Bench levels

Level	Who may do it	Typical repairs
LEVEL 1	Any trained student	Software fixes, single keys
LEVEL 2	Student with supervisor sign-off on this repair	Keyboard deck, battery, Wi-Fi, touchpad, speakers, I/O board, bezel, panel, cables, camera, hinges
LEVEL 3	Lead tech or adult supervisor only	Liquid damage, system board swap, heat sink

Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under HP warranty or an accidental-damage plan. A warranty claim may cost less than a part, and opening a unit can affect warranty coverage.
3. ☐ Photograph all four sides of the device before opening it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

Safety rules

- **Power off, unplug, remove any SD card and cables,** then disconnect the battery before any other work inside.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** wear a grounded wrist strap on an ESD mat. Touch boards only by the edges.
- **Ribbon cables:** unlock every ZIF latch before sliding a ribbon out. A locked ribbon tears.
- **Glass:** wear safety glasses when removing a cracked screen.
- **Before power-on:** check for loose screws inside. Shake gently and listen. A loose screw can short the board.

Tools and supplies

Item	Use	Notes
Phillips #0 and #1 screwdrivers	All screws	Most are M2.0 and M2.5 Phillips
Plastic spudger, opening picks, plastic tweezers	Clips, latches, ribbons	Only plastic touches connectors
ESD mat + wrist strap	Static protection	
Magnetic screw mat	Keeping screws in order	Screws are often different lengths
90%+ isopropyl alcohol + soft brush	Cleaning contacts, spill residue	
Kapton tape	Re-taping cables and connectors	
Recovery USB or SD card, 8 GB+	ChromeOS recovery	Chromebook Recovery Utility
Known-good USB-C charger	Testing power and charging	
Safety glasses	Cracked-screen removal	
Fire-safe battery bin or sand bucket	Damaged batteries	Required at every bench

2 Know the machine

Specs that matter for repair

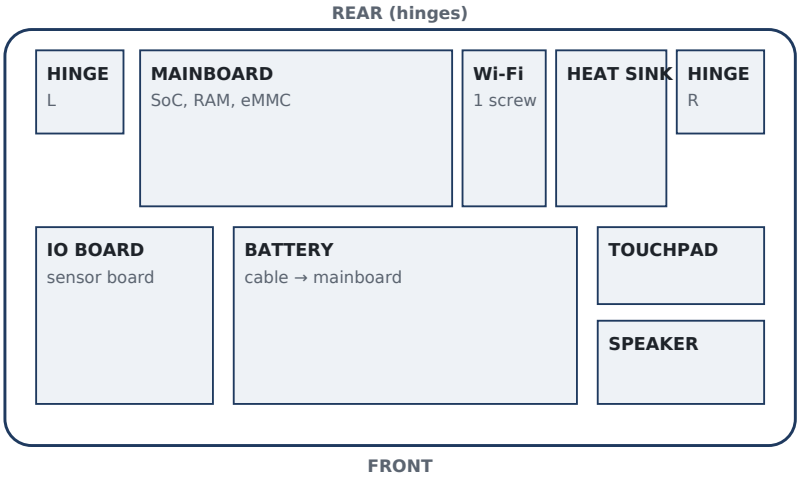
Item	11 G9 EE	Why it matters
Processor	Intel Celeron N4500 / N5100	Soldered
Memory / storage	4 or 8 GB RAM; 32 or 64 GB eMMC	Soldered
Display	11.6" HD (eDP)	29.5 cm panel
Battery	2-cell, 47 Wh class	Shared HP 11 battery family
Access	Keyboard deck from the top, held by 4 bottom screws	No bottom panel
Wireless	Wi-Fi card on a bracket, 1 screw	Replaceable
Sensors	Sensor board with its own cable	Replaceable

How it comes apart

Unlike most Chromebooks, this HP opens from the top. Screws on the bottom hold the keyboard deck (keyboard, palm rest and touchpad) to the bottom cover. Lift the deck off and the battery, boards, speakers and Wi-Fi are underneath. The display has its own bezel, panel, back cover, camera and hinges.

Inside the base

With the keyboard deck lifted off, you see the parts below. It's a rough map; correct it on your first teardown.



ILLUSTRATIVE — block positions are approximate. Confirm on your first teardown.

11 G9 EE parts list

HP spare part numbers from the official Maintenance and Service Guide. Look them up on partsurfer.hp.com before ordering; HP revises parts.

Part	Part number	Repair
Keyboard/top cover	by serial	R2
Battery / shield	L75783-005 / L98587-001	R3
Wi-Fi bracket	M45807-001	R5
Touchpad bracket / Cable Kit	L89791-001 / M44247-001	R6
Speakers	by serial	R7
Display bezel	L89773-001	R9
Panel	by serial	R10
Hinges	by serial	R13
I/O bracket	M45808-001	R17
System board	by serial	R17

3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through "try first" before ordering parts.

Symptom	Try first	Then	Part likely needed
Won't turn on, no lights	Known-good charger 30 min → S1	R4 → R3	Battery, or system board
Not charging	Other USB-C port and charger → S1	R3 → Sec. 6 if a port is dead	Battery or system board
Battery drains fast / low health in S2	S2 battery test	R3	Battery
Swollen battery / bulging case	STOP — Level 3	R3 by lead tech	Battery
Black screen, power light on	External display via USB-C; S1	R4 → R11 → R10	Display cable or panel
Flicker or lines that change with lid angle	S1	R11	Display cable
Cracked or blotchy screen	—	R10	LCD panel
Bezel loose or broken	Reseat clips	R9	Bezel
Lid floppy, hinge cracked	Check hinge screws	R13	Hinge(s)
One key missing or broken	—	R1	Keycap
Several keys or a row dead	S2 keyboard test; S1; S4	R4 → R14	Keyboard / top cover
Keys type wrong symbols	S4	S4	None
Touchpad dead or erratic	S1; check touchpad settings	R4 → R6	Touchpad or its cable
USB-A / SD card / audio dead	Try another device in the port; S1	R8	I/O board or its cable
Camera not detected	Privacy shutter open? S1	R12	Camera module
No sound (headphones work)	Volume, output; S1; S3	R7	Speakers
Wi-Fi missing or weak	S1 → S3	R5	Wi-Fi card or antennas
Runs hot, fan loud, shuts down	Check vents for lint; S2	R15	Fan or heat sink
Liquid spill	Power off NOW, unplug	R16	Depends on damage
"ChromeOS is missing or damaged"	S3 recovery	R17	System board (storage)
Cracked bottom or lid shell	—	R2	Cover

4 Software repairs

No tools needed. Always do S1 before opening a device.

S1 Hard reset (EC reset) LEVEL 1

Fixes	Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad
Time	2 minutes
Tools	Known-good USB-C charger
Data loss	None

A hard reset restarts the embedded controller (power, keyboard and charging chip). It erases nothing.

Procedure

1. ☐ Unplug the charger and any USB devices. Remove any SD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Refresh** (circular arrow), then tap **Power**. Release Refresh when it starts.
4. ☐ If nothing happens: plug in a known-good charger, wait 30 minutes, and repeat with the charger connected. Try the other USB-C port too.

Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

Tip. After a battery has been disconnected, plug in a charger before deciding the device is dead. Many Chromebooks won't wake on the power button alone.

S2 Built-in hardware diagnostics LEVEL 1

Fixes	Confirms the fault before ordering a part: battery, CPU, memory, keyboard, touchpad, Wi-Fi
Time	5–15 minutes
Tools	None
Data loss	None

Procedure

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the Launcher, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key; test the touchpad and touchscreen.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

S3 Powerwash and USB recovery — LEVEL 1

Fixes	Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", Wi-Fi missing after an update
Time	Powerwash 10 min; recovery 20–40 min
Tools	Recovery USB or SD card (8 GB+)
Data loss	**Yes.** Erases all local files (Downloads).

District-managed devices. Enrolled devices should force re-enrollment on their first Wi-Fi connection after a wipe. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi reaches enrollment. Never enter Developer Mode on a district device.

Procedure

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates.
2. ☐ **Powerwash:** sign out, press Ctrl + Alt + Shift + R, choose Restart, then Powerwash. (May be blocked by policy; use recovery instead.)
3. ☐ **Make recovery media:** on a working computer, open the Chromebook Recovery Utility, choose the **HP Chromebook 11 G9 EE** from its list (or search the model), insert an 8 GB+ drive, and click Create now.
4. ☐ Hold **Esc + Refresh**, tap **Power**, and release when the recovery screen appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm it re-enrolls to the KPS domain.

Verify before closing the ticket

- Boots to the enrollment or KPS sign-in screen.
- Shows as enrolled in the Admin console.
- Recovery can't write to storage: storage has failed. That's a system board repair or warranty claim.

S4 Keyboard layout and international keyboard fix — LEVEL 1

Fixes	Keys type the wrong symbols (@ and " swapped, accents appear, Z and Y swapped); a replacement keyboard with a different layout
Time	5 minutes
Tools	None
Data loss	None

ChromeOS can have several keyboard layouts installed. Students often switch layouts by accident with **Ctrl + Shift + Space**, or have **US International** on, which turns ' and " into accent keys. The physical keyboard is fine.

Procedure

1. ☐ Look at the bottom-right shelf. A layout code (US, INTL, UK, ES...) means more than one layout is installed.
2. ☐ Open **Settings > Device > Keyboard and inputs > Input** (on older versions: Settings > Advanced > Languages and inputs > Inputs).
3. ☐ Set **English (US)** as the active input. Remove any layouts the student doesn't need, especially US International, unless a teacher asked for it.
4. ☐ Test in a text box: Shift + 2 should type @ and Shift + ' should type ".
5. ☐ If a **replacement keyboard** has a different physical layout (UK keyboard on a US unit, for example), order the correct US part. Changing the software layout won't fix printed key labels.
6. ☐ If the wrong layout keeps coming back for everyone who signs in, it's set by district policy. Tell the lead tech; don't change it on the device.

Verify before closing the ticket

- Every key types its printed character in the S2 keyboard tester and in a text box.
- Only the needed layouts remain installed.

5 Hardware repairs

Every internal repair on this model starts with the keyboard deck (R2) and disconnecting the battery. Screw counts and sizes come from HP's guide.

Golden rules. Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

R1 Single keycap replacement — LEVEL 1

Fixes	Missing, cracked or loose key
Part	Individual key (ask the manufacturer or harvest from a scrapped keyboard)
Time	5 minutes
Tools	Plastic pick, tweezers

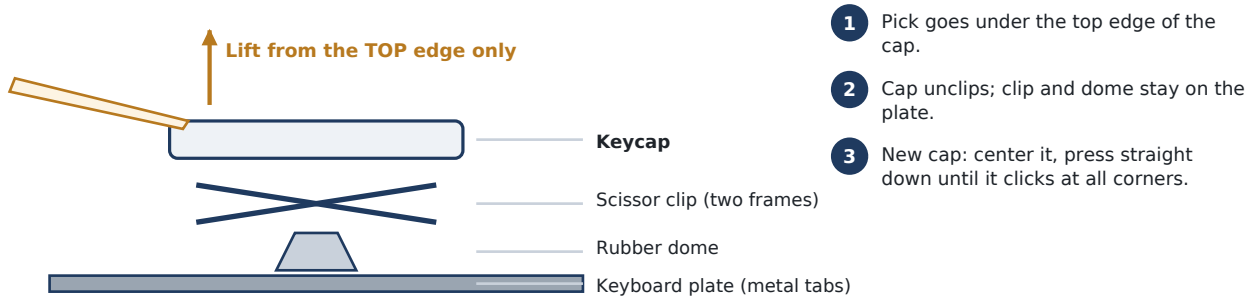


Illustration: side view of one key. Not to scale.

Procedure

1. ☐ Power off. Open the lid flat on the mat.
2. ☐ Slide a plastic pick under the TOP edge of the damaged key and lift until it unclips.
3. ☐ Check the scissor clip and rubber dome under it. A broken clip must be replaced too.
4. ☐ Fit a clip: hook its pins under the metal tabs on the keyboard plate, then press flat.
5. ☐ Center the new key over the clip and press straight down until it clicks at every corner.
6. ☐ Wide keys (Space, Shift, Enter): seat the metal stabilizer wire in its holders first.

Verify before closing the ticket

- Key clicks evenly and springs back.
- Key registers in the S2 keyboard tester.

R2 Keyboard / top cover removal (the access cover) LEVEL 2

Fixes	Dead keyboard, cracked palm rest; the first step for every internal repair on this model
Part	Keyboard/top cover (by country; look up by serial)
Time	15 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, screw mat

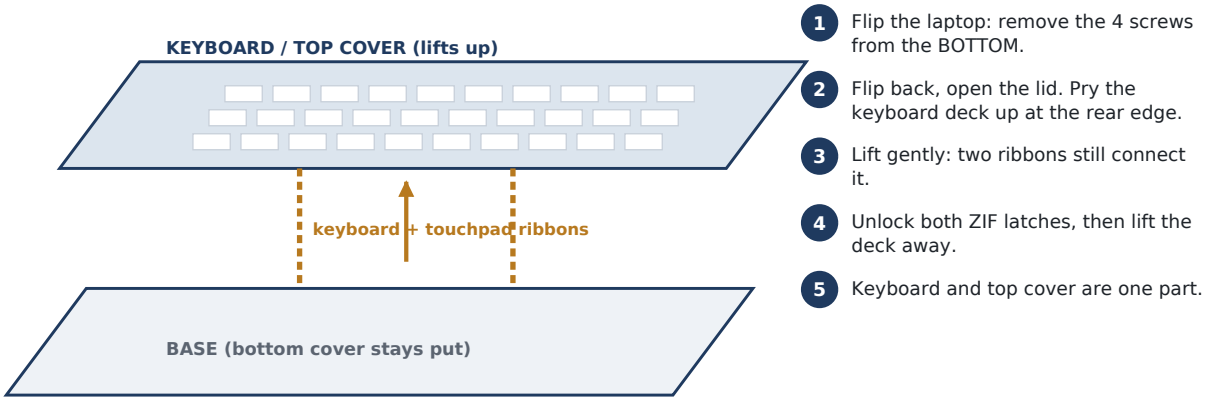


Illustration: on this model you go in from the top. The keyboard deck is the access cover.

On this model you go in from the top. There is no removable bottom panel. The keyboard deck (keyboard + palm rest + touchpad) is the access cover, and the keyboard is built into it. "Keyboard replacement" and "top assembly" are the same part here.

Procedure

1. ☐ Shut down fully, unplug, remove any SD card.
2. ☐ Close the lid and turn the computer upside down on the mat.
3. ☐ Remove the 4 Phillips M2.0×7.1 screws (11MK value — confirm) that hold the keyboard/top cover to the bottom cover. Note which holes take the longer screws.
4. ☐ Turn it right side up and open the lid as far as it goes.
5. ☐ Starting at the rear edge (near the hinges), slide a plastic pick into the seam and release the clips along the back, then the sides.
6. ☐ Lift the rear of the keyboard deck slowly, like opening a book toward you. Don't pull: the keyboard and touchpad ribbons are still connected.
7. ☐ Unlock the ZIF latches and slide out the touchpad ribbon (and keyboard ribbon). Lift the deck away.
8. ☐ **Disconnect the battery now** (see the battery repair, first steps) before touching anything else.
9. ☐ To refit: reconnect the ribbons and lock the latches, lay the deck on, press around the edges until every clip snaps, then refit the bottom screws.

Verify before closing the ticket

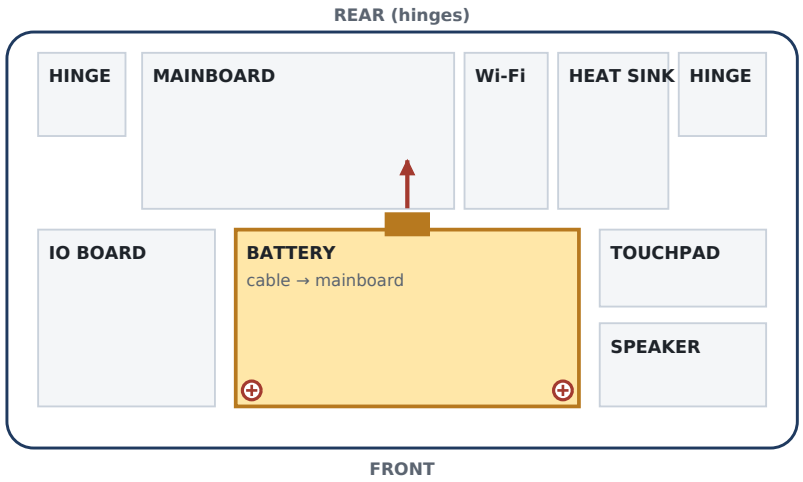
- Keyboard and touchpad pass the S2 tests after refitting.

- No gaps at the seam; every bottom screw refitted.

First-teardown notes — fill in the first time you do this repair	
Which holes take the long screws?	_____
Where the first clip released	_____

R3 Battery replacement — LEVEL 2

Fixes	Won't hold charge, low health in S2, swollen battery
Part	Battery L75783-005 (2-cell, 47 Wh) — confirm by serial; battery shield L98587-001
Time	15–20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap



1. Unplug the battery cable from the mainboard first (red arrow).
2. Pry at the plug body, never the wires.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

Illustration: base interior, battery highlighted. Screw positions illustrative.

Swollen battery = Level 3. Don't press or pry. The lead tech handles it; the old cell goes straight to the fire-safe bin.

Procedure

1. ☐ Remove the keyboard deck (top cover repair above).
2. ☐ Lift the tape or shielding over the battery connector, then disconnect the battery cable from the system board. Pry at the connector body.
3. ☐ Remove the battery screws (count them; HP's 11MK guide shows them in its illustration) that hold the battery to the bottom cover.
4. ☐ Lift the battery out. Never bend it.
5. ☐ Fit the new battery, refit its screws, reconnect the cable, and re-tape the connector.
6. ☐ Refit the keyboard deck. Plug in the charger before powering on.

Verify before closing the ticket

- Charging light comes on; battery percentage shows.

- S2 battery health near 100% on a new battery.
- Old battery goes to battery recycling, never the trash.

Battery shield. Remove the battery shield and free the speaker cable before lifting the battery, as HP's guide shows.

R4 Reseating internal connectors

LEVEL 2

Fixes	Dead keyboard, touchpad or speakers; black screen; dead ports; no power after a drop
Part	None, unless a cable is damaged
Time	15–25 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol

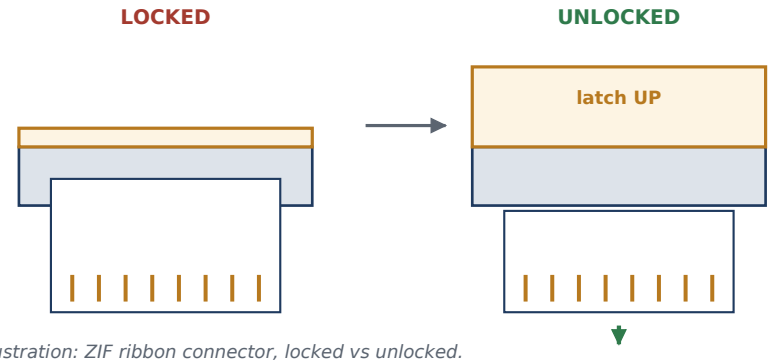


Illustration: ZIF ribbon connector, locked vs unlocked.

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Connector	Type	How to release
Keyboard ribbon	ZIF ribbon	Unlock the latch; slide out
Touchpad ribbon	ZIF ribbon	Unlock the latch; slide out
Battery cable	Board-to-wire	Lift tape; pry at the connector body
Display (eDP) cable	Locking connector	Lift tape and the locking bar, then pull straight
Sensor board cables	ZIF ribbons	Unlock both ends first
Speaker cable	Board-to-wire	Pry at the plug body
Wi-Fi antennas (2)	Snap-on coax	Lift straight up; #1 white = Main, #2 black = Aux

Procedure

1. ☐ Open the device and disconnect the battery.
2. ☐ Hold Power for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ For each connector related to the symptom: photograph it, unlock and release it, and inspect it for bent pins, burn marks or creases.
4. ☐ Clean dirty contacts with a brush dampened in isopropyl alcohol. Let them dry 5 minutes.
5. ☐ Reconnect each one. Ribbons must go in straight and all the way before you close the latch.

6. ☐ Reconnect the battery last. Plug in and test **before** closing up.

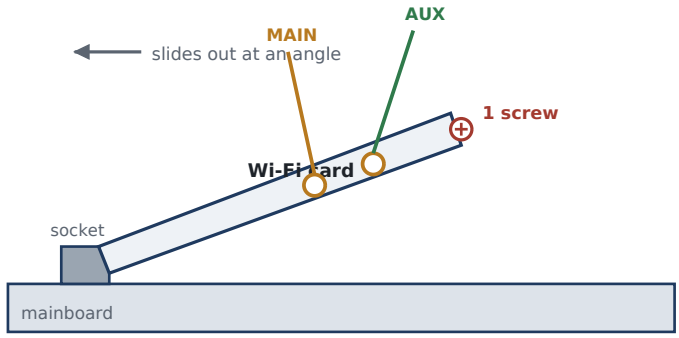
Verify before closing the ticket

- Symptom gone in a full test, not just at boot.
- Creased or torn ribbons replaced.

R5 Wi-Fi card reseat or replacement

LEVEL 2

Fixes	No Wi-Fi or Bluetooth, weak signal, Wi-Fi gone after a drop
Part	Wi-Fi card (look up by serial)
Time	15 minutes
Tools	Phillips driver, plastic tweezers



1. Photograph which antenna goes on which terminal.
2. Lift both antenna plugs straight up with plastic tweezers.
3. Remove the 1 screw. The card springs up at an angle.
4. Slide it out of the socket. Refit in reverse.

Illustration: M.2 Wi-Fi card, side view. Colors of antenna leads vary.

Procedure

1. ☐ Remove the keyboard deck and disconnect the battery.
2. ☐ Find the Wi-Fi card on the system board.
3. ☐ Note which antenna is on which terminal: #1 (white) goes to Main, #2 (black) to Aux. Lift both straight off.
4. ☐ Remove the 1 Phillips M2.0×3.5 screw and the Wi-Fi bracket (M45807-001); remove the clear shield first.
5. ☐ The card tilts up. Slide it out of the socket at an angle.
6. ☐ Refit at an angle, press down, refit the screw, snap on both antennas, reconnect the battery and test.

Verify before closing the ticket

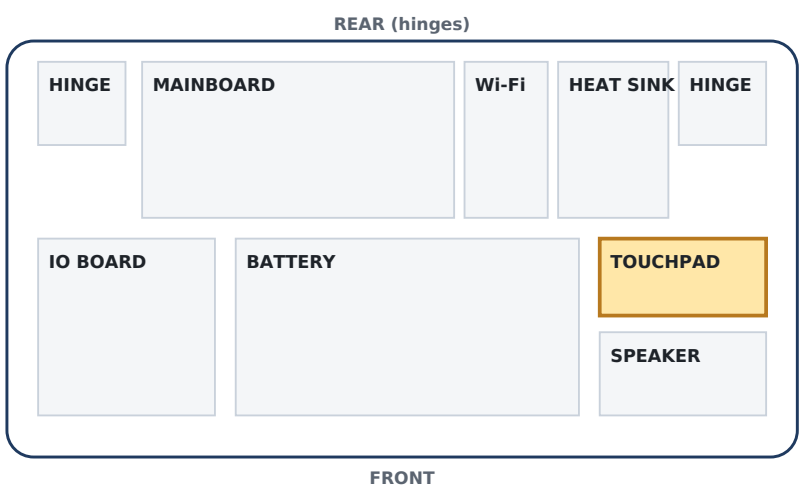
- Wi-Fi connects and the S2 connectivity tests pass.
- Bluetooth finds a device.
- Weak signal after a lid repair usually means a pinched antenna lead.

Use an approved card. HP warns that an unauthorized card can make the system unresponsive. Use the HP spare part, or move the card from a donor of the same model.

R6 Touchpad and touchpad cable replacement

LEVEL 2

Fixes	Touchpad dead, erratic, cracked or won't click (keyboard works)
Part	Touchpad (by serial); bracket L89791-001; cable in Cable Kit M44247-001
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape



1. Battery disconnected first.
2. Unlock the ZIF latch before sliding the ribbon out.
3. Lift tape or Mylar; remove the screws.
4. Conductive tape must touch the pad on reassembly.

Illustration: base interior, touchpad highlighted. Position illustrative.

Check first: touchpad turned on in Settings, S1 done, and the cable reseated.

Procedure

1. ☐ Remove the keyboard deck and disconnect the battery. The touchpad is in the underside of the deck.
2. ☐ Peel back the pad or tape over the touchpad cable, unlock its ZIF latch at the touchpad, and remove the cable.
3. ☐ Release any shielding along the top edge of the touchpad.
4. ☐ Remove the touchpad bracket screws (count them) that hold the touchpad and its bracket.
5. ☐ Remove the bracket, then lift out the touchpad.
6. ☐ Fit the new touchpad and bracket, refit the screws, restick the shielding, and reconnect the cable.

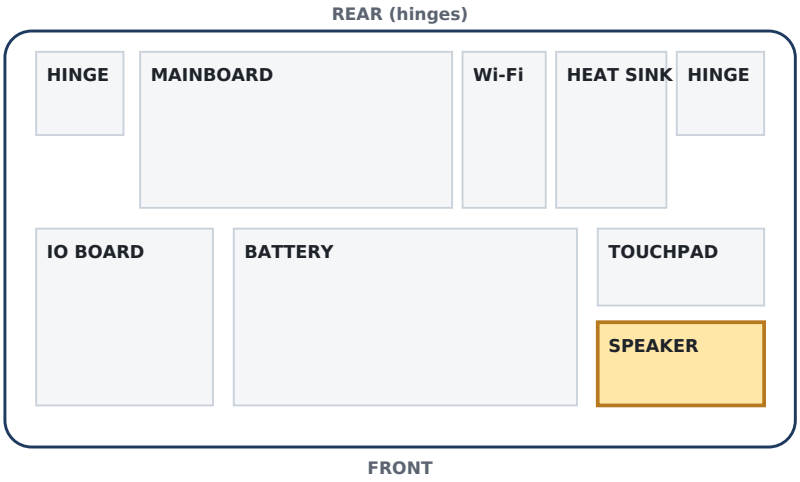
Verify before closing the ticket

- Cursor moves, clicks and two-finger scrolls.
- Click works at all four corners.
- Passes the S2 touchpad test.

R7 Speaker replacement

LEVEL 2

Fixes	No sound or crackling from the speakers, with headphones working
Part	Speakers (by serial)
Time	20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers



1. Battery disconnected.
2. Unplug the speaker cable at the board.
3. Unroute the cable, noting each clip.
4. Keep the rubber isolators: they stop rattle.

Illustration: base interior, speaker highlighted. Position illustrative.

Check first: if headphones also have no sound, the fault is software or the system board, not the speakers.

Procedure

1. ☐ Remove the keyboard deck and disconnect the battery.
2. ☐ Lift any tab and disconnect the speaker cable from the system board.
3. ☐ Release the speaker cable from its routing clips.
4. ☐ Remove the speaker screws (count them).
5. ☐ Lift out the speakers with their rubber isolators.
6. ☐ Fit the new speakers (isolators in place), route the cable through every clip, and reconnect.

Verify before closing the ticket

- Clear sound from both sides with no rattle at full volume.
- Headphone jack still works.

R8 I/O (USB) daughterboard replacement

LEVEL 2

Fixes	Dead Sensors with the system board working
Part	Sensor board and cable (by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers

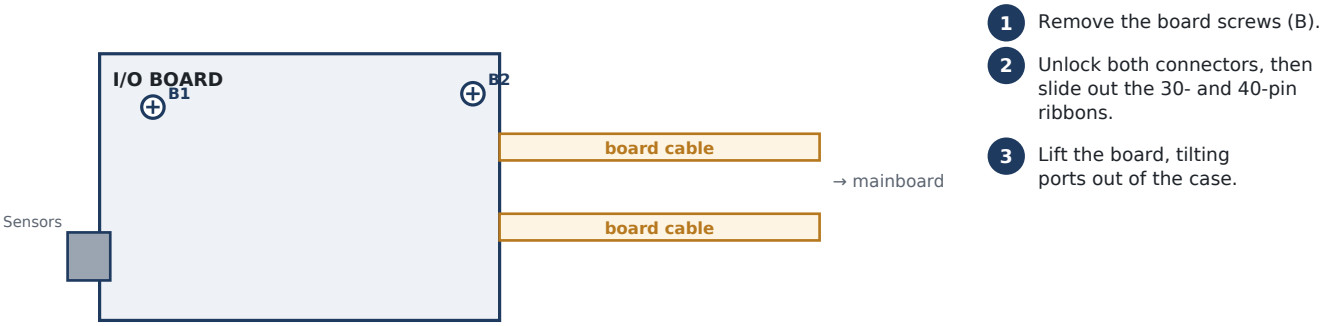


Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.

Procedure

1. ☐ Remove the keyboard deck and disconnect the battery.
2. ☐ Release the shielding and unlock the ZIF connectors at both ends of the sensor board cables; remove the cables.
3. ☐ Check whether a speaker or bracket covers the board; move it aside if so.
4. ☐ Remove the sensor board screws (count them).
5. ☐ Remove the I/O bracket, then lift out the board.
6. ☐ Fit the new board and bracket, refit the screws, refit the cables with the latches locked, reconnect the battery and test every port.

Verify before closing the ticket

- Every port on the board works (flash drive, charger, SD card, headphones as fitted).
- Bracket and all screws refitted.

R9 Display bezel replacement

LEVEL 2

Fixes	Cracked, loose or missing screen bezel; the first step for panel, camera and display-cable repairs
Part	Display bezel L89773-001 (includes adhesive)
Time	10 minutes
Tools	Plastic picks, Phillips driver

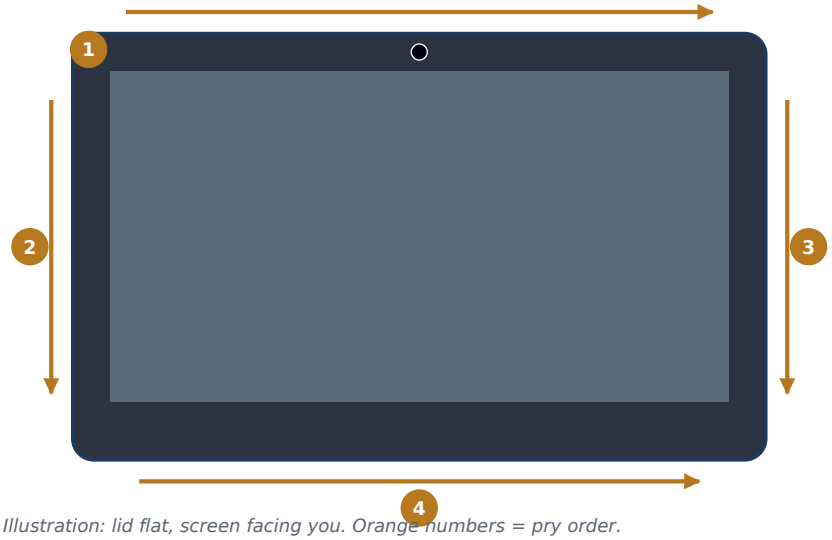


Illustration: lid flat, screen facing you. Orange numbers = pry order.

1. This bezel is glued as well as clipped.
2. Warm the edge gently (hair dryer on low) if it resists.
3. Slide a thin pick along the adhesive; don't bend the bezel.
4. Fit new adhesive strips on reassembly.
5. Don't pull on the camera window.

Procedure

1. ☐ Shut down and unplug. Open the lid flat on the mat.
2. ☐ Release the top edge of the bezel with a plastic pick.
3. ☐ Work down the left and right sides.
4. ☐ Release the bottom edge last. It has double-sided adhesive; go slowly.
5. ☐ To fit: line up the camera window, press around the edge until every clip seats.

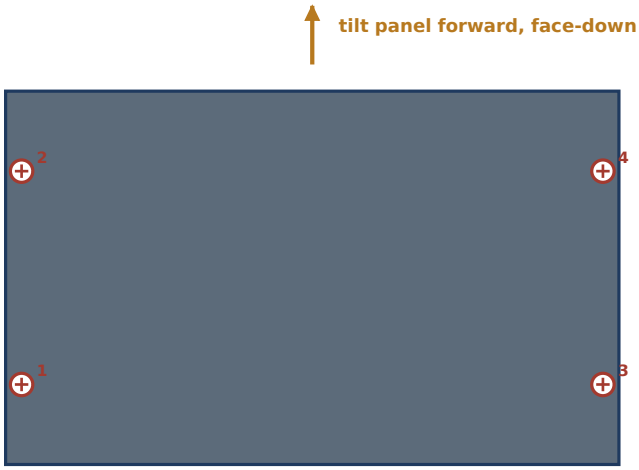
Verify before closing the ticket

- Bezel sits flush all round.
- Camera window lined up; privacy shutter slides freely.
- Lid closes fully.

R10 LCD panel replacement

LEVEL 2

Fixes	Cracked screen, dead pixels, blotches, black screen not fixed by the display cable
Part	Panel 11.6" HD (by serial)
Time	25–35 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, safety glasses, Kapton tape



1. Remove the 4 panel screws (sides of the panel).
2. Lay the panel face-down on the keyboard (cloth under it).
3. On the back: peel the tape over the connector, open the clip, pull the cable straight out.
4. Connect the new panel and TEST before screwing it down.

Illustration: lid with bezel removed. Screw positions are typical. Count them on yours.

Cracked glass. Wear safety glasses. Tape across a shattered panel before handling it.

Test first: if an external USB-C display works, the system board is fine and the fault is the panel or its cable.

Procedure

1. ☐ Remove the keyboard deck and disconnect the battery.
2. ☐ Remove the bezel.
3. ☐ Remove the 6 Phillips M2.0×2.9 screws (panel bracket / hinge brackets) that hold the panel and bracket to the back cover.
4. ☐ Lift the top edge and swing the panel forward until it rests face down on the keyboard (cloth underneath).
5. ☐ Release the adhesive support strip, lift the locking bar, and disconnect the panel cable.
6. ☐ Connect the new panel, close the locking bar, restick the strip, and test the picture **before** screwing the panel down.
7. ☐ Refit the screws, bezel and keyboard deck.

Verify before closing the ticket

- Full image: no lines, dead pixels or uneven backlight.
- Brightness works across its whole range.
- Image steady through full lid travel.

R11 Display (eDP) cable replacement

LEVEL 2

Fixes	Flicker, lines or black screen that changes with lid angle; external display works
Part	Display cable (by serial)
Time	40–50 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape

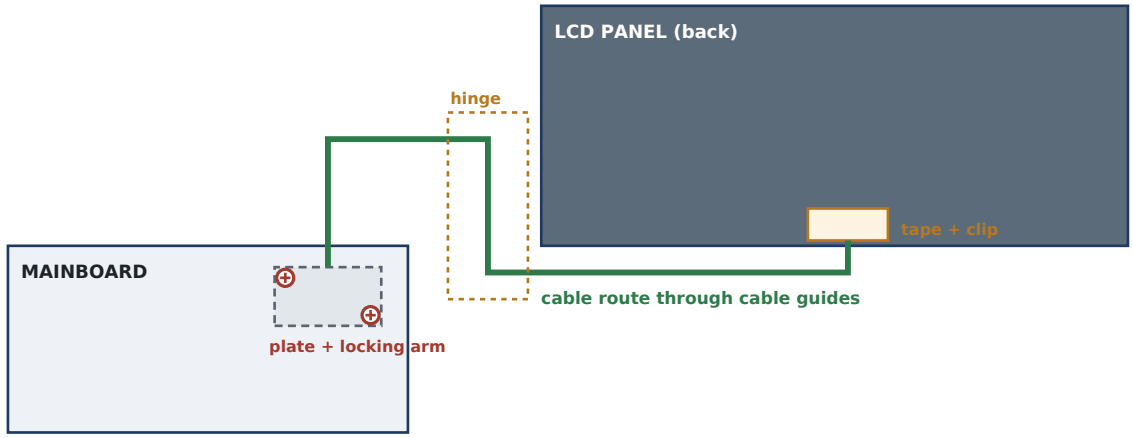


Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.

Procedure

1. ☐ Remove the keyboard deck, disconnect the battery, and disconnect the panel cable at the system board.
2. ☐ Remove the bezel and panel (panel repair steps).
3. ☐ Photograph the route, then free the cable from the retention clips and routing channel.
4. ☐ Peel it off the back cover (double-sided adhesive).
5. ☐ Route the new cable exactly along the old path, through every clip, and reconnect both ends.
6. ☐ Test at every lid angle before reassembly.

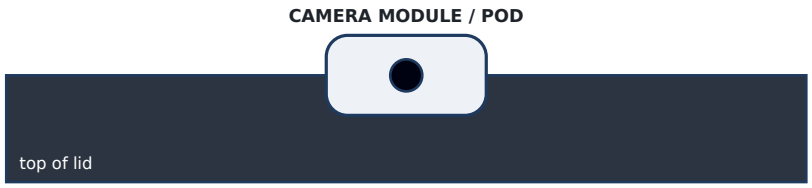
Verify before closing the ticket

- Steady image at every angle: open and close the lid 20 times while watching.
- Camera still works if it shares the cable.

R12 Camera module replacement

LEVEL 2

Fixes	Camera not detected, black image, broken lens
Part	Webcam/microphone module (adhesive); cable in Cable Kit M44247-001
Time	15–20 minutes
Tools	Plastic picks, plastic tweezers



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

Check first: privacy shutter open? Camera works after S1 and a powerwash (S3)?

Procedure

1. ☐ Remove the keyboard deck and disconnect the battery.
2. ☐ Remove the bezel.
3. ☐ The camera module is held by double-sided adhesive at the top of the lid. Slide a thin plastic pick under it.
4. ☐ Release the ZIF connector and disconnect the camera cable.
5. ☐ Fit the new module (it comes with adhesive), reconnect, and test before refitting the bezel.

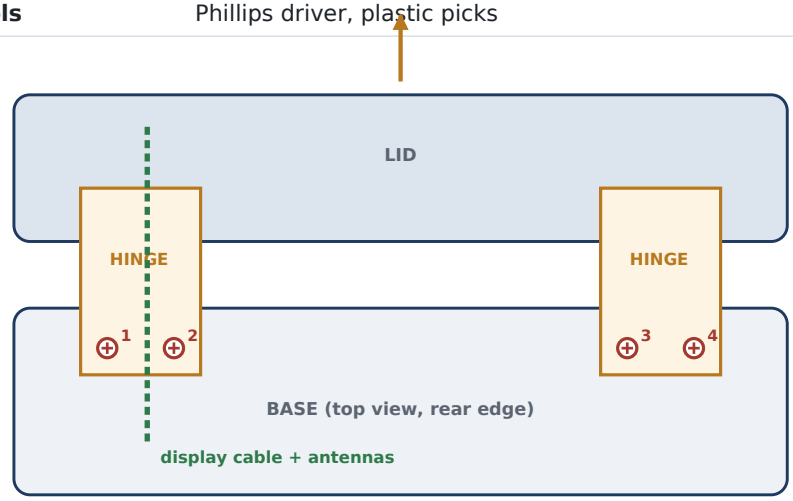
Verify before closing the ticket

- Camera app shows a clear image.
- Privacy shutter fully covers the lens when closed.
- Works in a Google Meet test call.

R13 Hinge replacement

LEVEL 2

Fixes	Floppy lid, cracked hinge, grinding, lid won't stay up
Part	Hinges (by serial)
Time	30–45 minutes
Tools	Phillips driver, plastic picks



1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Remove the hinge screws (count them) (hinges to bottom cover).
4. Support the lid, then lift it away from the base.

Illustration: hinge area seen from above with the D cover off. Positions approximate.

Check first: tighten the hinge screws. A screw that spins without tightening means a stripped post: that's a cover replacement.

Procedure

1. ☐ Remove the keyboard deck and disconnect the battery.
2. ☐ Disconnect the Wi-Fi antennas and the display and camera cables at the system board.
3. ☐ Remove the hinge screws (count them) that hold the hinges to the bottom cover.
4. ☐ Open the display fully, slide the hinges out, and lift the display off the base.
5. ☐ To replace the hinges on the lid: remove the bezel, then the back-cover screws that hold the back cover to the hinges.
6. ☐ Fit the new hinges (left and right are mirror images), refit every screw, route the cables back through the hinges, and reconnect.

Verify before closing the ticket

- Lid holds any angle through its full travel.
- Both hinges feel equally stiff.
- Display steady and Wi-Fi strength normal (no pinched cables).

First-teardown notes — fill in the first time you do this repair

Which hinge the display cable passes through _____

R14 Keyboard (top cover) replacement

LEVEL 2

Fixes	Several dead keys, a dead row, liquid-damaged keyboard, cracked palm rest
Part	Keyboard/top cover (by country; look up by serial)
Time	60–90 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap, screw mat, Kapton tape



Illustration: what moves to the new top assembly. Check what comes pre-fitted first.

Procedure

1. ☐ Remove the keyboard deck (top cover repair). On this model that IS the keyboard replacement.
2. ☐ Compare old and new decks. Move anything the new one lacks: touchpad and bracket, rubber bumpers, Mylar, tape.
3. ☐ Reconnect the keyboard and touchpad ribbons to the system board and lock the latches.
4. ☐ Fit the new deck and refit the bottom screws.

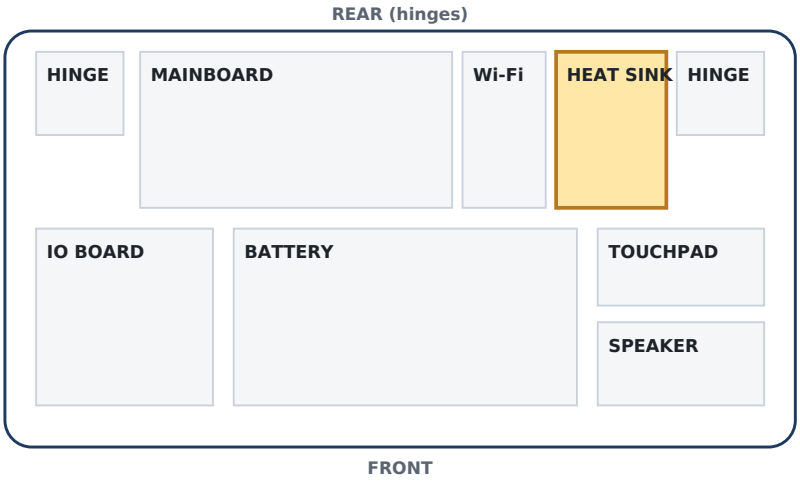
Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, ports, speakers and Wi-Fi all work.
- No screws left over.

R15 Heat sink replacement

LEVEL 3

Fixes	Overheating, thermal shutdowns, loud or rattling fan
Part	Heat sink (thermal material included in kit)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol, lint-free cloth



1. Battery disconnected.
2. Loosen the screws in the numbered order printed on the heat sink.
3. Clean old thermal paste with alcohol.
4. New thermal material, then tighten in order.

Illustration: base interior, cooling highlighted. Position illustrative.

Thermal material. Clean old thermal material off the processor and heat sink every time, and fit new material. Never run the board without the heat sink.

Procedure

1. ☐ Remove the system board.
2. ☐ Remove the heat sink screws and lift it off.
3. ☐ Clean old thermal material from the processor and heat sink every time.
4. ☐ Apply the replacement thermal material from the kit and refit.

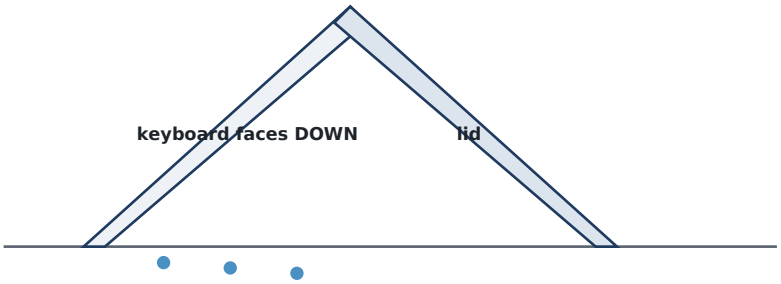
Verify before closing the ticket

- Device runs a 10-minute video without shutting down.
- Fan (if fitted) spins up under load and is quiet at idle.

R16 Liquid spill response

LEVEL 3

Fixes	Any liquid on or in the device
Part	Often none; sometimes keyboard/top cover or system board
Time	10 min first response; 24–48 h drying
Tools	90%+ isopropyl alcohol, soft brush, lint-free cloths



1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

Illustration: drain position after a spill.

Spill-resistant keyboards resist drips, not a full cup. Speed matters more than anything. HP's keyboard deck is a drip tray: liquid often stops above the system board.

Do not power on to test. Current through liquid causes the corrosion that kills boards.

Procedure

1. ☐ Hold Power for 10 seconds to force off. Unplug.
2. ☐ Tent the device upside down (lid at 90°, keyboard down) to drain.
3. ☐ Open it and disconnect the battery as soon as possible.
4. ☐ Blot visible liquid; don't wipe it around.
5. ☐ Sugary or sticky spills: clean affected areas with a soft brush and isopropyl alcohol.
6. ☐ Dry open, battery disconnected, for 24 hours minimum (48 if soaked). No hair dryer, no rice.
7. ☐ Inspect for white or green corrosion; clean connectors.
8. ☐ Reconnect the battery, plug in, and run S2 fully.

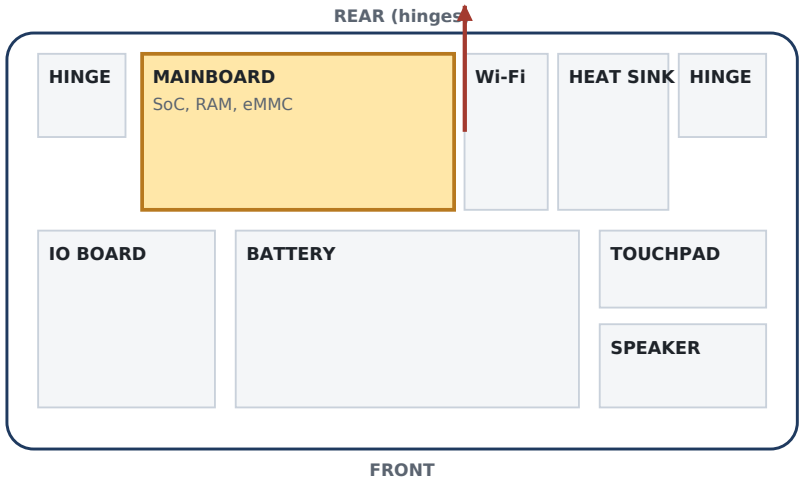
Verify before closing the ticket

- All S2 tests pass.
- Sticky or dead keys: keyboard repair.
- Corrosion on the system board: Section 6.

R17 System board replacement (lead tech)

LEVEL 3

Fixes	Dead USB-C ports, no power with a good battery, failed storage, RAM errors, failed recovery
Part	System board (by serial)
Time	45–90 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap



1. Battery out first (and the Wi-Fi card if it's socketed).
2. Unplug: battery, sensor board, display, webcam and speaker cables
3. Lift and slide out at an angle.
4. Lift by the edges onto the ESD mat.

Illustration: base interior, mainboard highlighted. Red arrow = lift side.

Identity and enrollment. Storage, serial number and enrollment live on the system board. The lead tech updates the asset record, deprovisions the old device in the Admin console if needed, and plans re-enrollment. Ask the manufacturer whether the new board ships with the serial set.

Confirm first with S1, S2, S3 and a connector reseat. Most "dead boards" are a battery, cable or connector.

Procedure

1. ☐ Remove the keyboard deck, disconnect the battery, and remove the Wi-Fi card.
2. ☐ Disconnect the battery, sensor board, display, webcam and speaker cables.
3. ☐ Remove the system board screws (count them).
4. ☐ Slide the board up and to the left at an angle, as HP's guide shows; note the I/O bracket (M45808-001) position.
5. ☐ Remove the heat sink and transfer it to the new board with fresh thermal material.
6. ☐ Fit the new board, reconnect everything, reconnect the battery, then run recovery (S3) if needed and enroll.

Verify before closing the ticket

- Boots, enrolls and passes all of S2.
- Asset record and Admin console show the correct serial number.

6 Out of scope: escalate

These need soldering or board-level work, or parts that aren't sold. Tag the device with the ticket and send it to the lead tech.

Fault	Why	Route
USB-C port loose or dead	Soldered to the system board	System board swap or HP warranty
Not charging with good battery and charger	Charging circuit on the system board	System board or HP warranty
Storage (eMMC) or RAM failure	Soldered	System board or HP warranty
Corrosion on the system board	Board-level work	System board or HP warranty
Burn marks, melted plastic, burnt smell	Fire hazard	Lead tech immediately
Any repair needing a soldering iron or hot air	Program policy	HP warranty or lead tech

HP support

- **HP Education support:** support.hp.com (search the model for warranty status and drivers)
- **HP parts lookup:** partsurfer.hp.com (enter the serial number for exact part numbers)
- **Service guide:** HP Chromebook 11MK G9 EE Maintenance and Service Guide

7 Sources

Checked October 2026. The procedures are the guide author's adaptation of these sources for a student bench; they are not manufacturer-issued instructions.

- HP, *Maintenance and Service Guide: HP Chromebook 11MK G9 Education Edition*, 76 pp. (viewed via manualslib.com/manual/2074931) — screw counts, part numbers, procedure order. Used as the reference for the 11 G9 EE.
- HP PartSurfer (partsurfer.hp.com).